Exploratory Data Analysis on a Cars Dataset

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Dataset: UCI Auto MPG

Source: UCI Machine Learning Repository — Auto MPG

1. Objective

This project inspects, cleans, summarizes and visualizes automobile data to identify patterns, relationships and potential outliers. A machine-learning model is not required.

2. Dataset Overview

The dataset had 398 rows before cleaning and 9 columns. Variables include MPG, cylinders, displacement, horsepower, weight, acceleration, model year, origin and car name.

3. Data Quality and Cleaning

Duplicate rows detected before cleaning: 0. Missing values before cleaning: horsepower: 6.

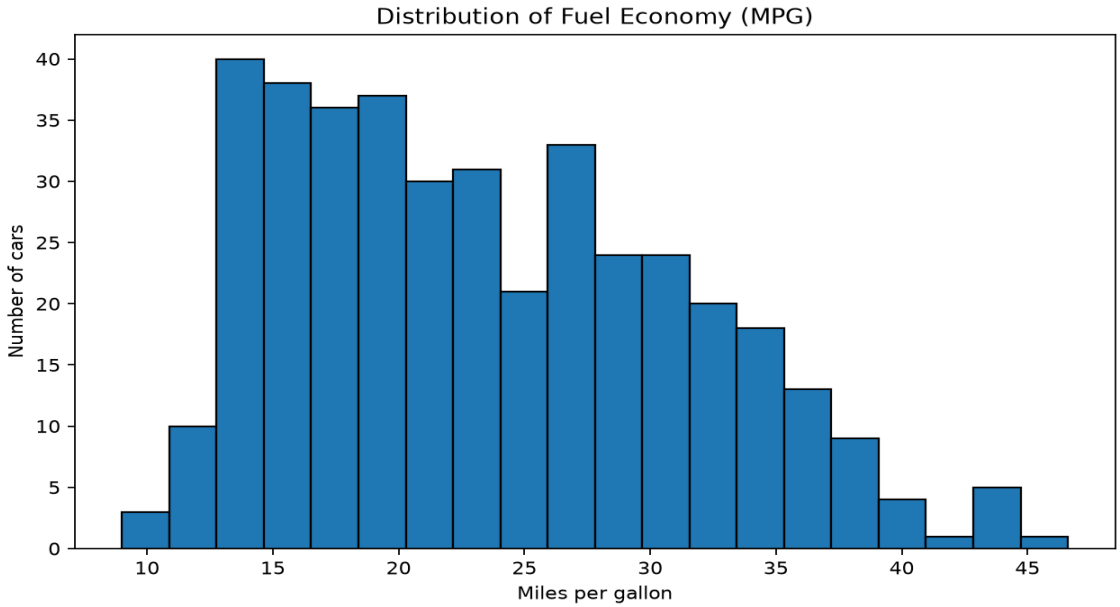
Rows before cleaning: 398; rows after cleaning: 398. Missing horsepower values were replaced with the median. Rows missing essential fields (MPG, weight, model year or origin) were removed. Potential outliers were flagged using the IQR method and retained rather than automatically deleted.

4. Descriptive Analysis

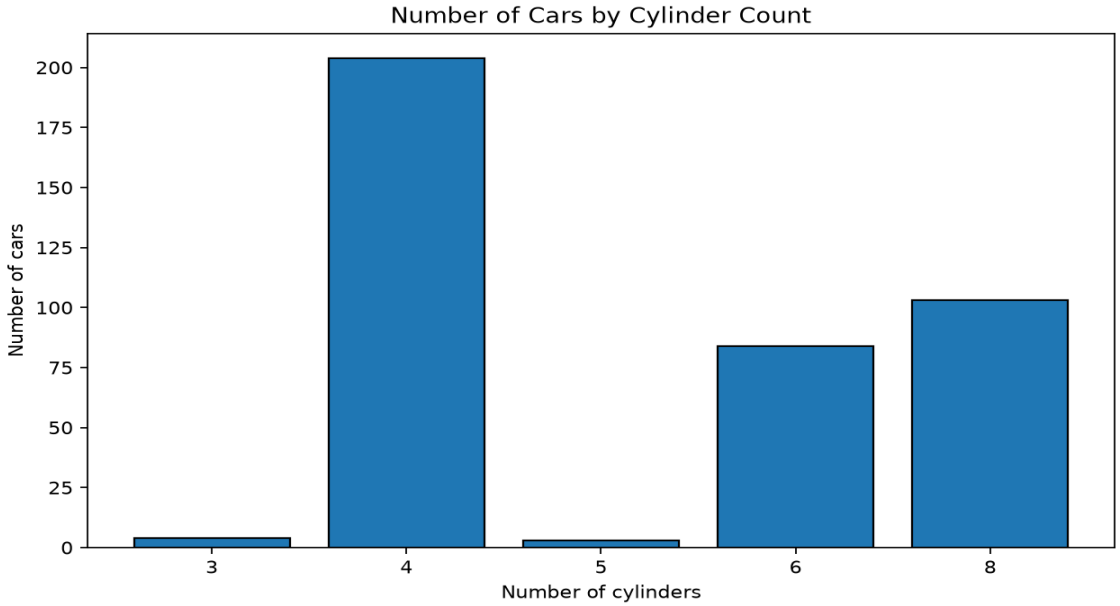
Mean MPG: 23.51; median MPG: 23.00; standard deviation of MPG: 7.82. Full descriptive statistics are included in the generated output folder.

5. Visual EDA

Each chart below is generated from the cleaned UCI dataset. Observations should be interpreted from the plotted evidence.

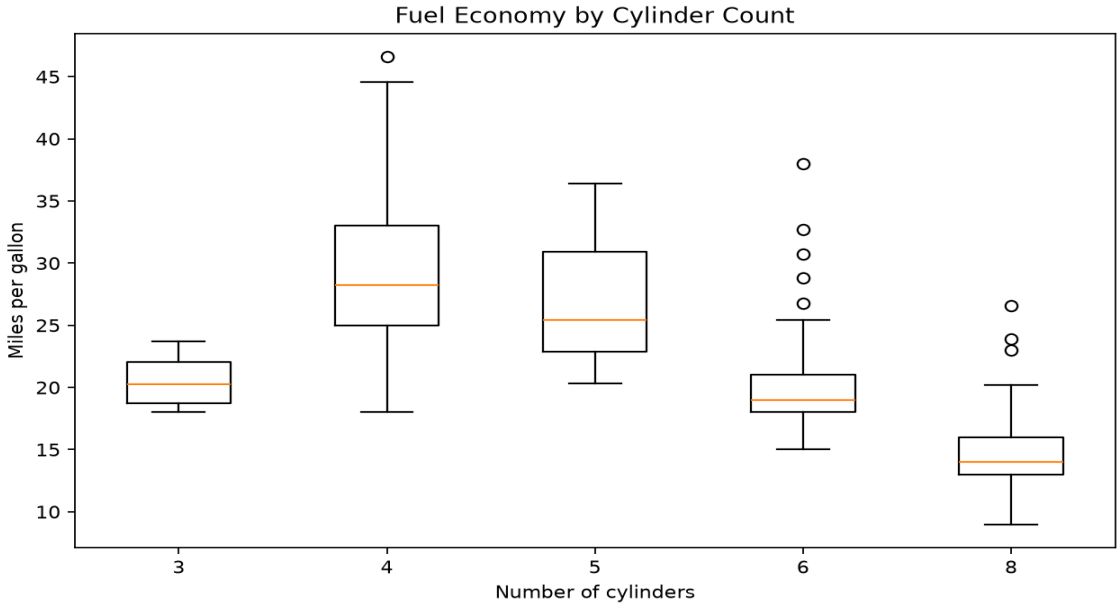


Observation: describe the distribution, spread and possible skew of MPG.

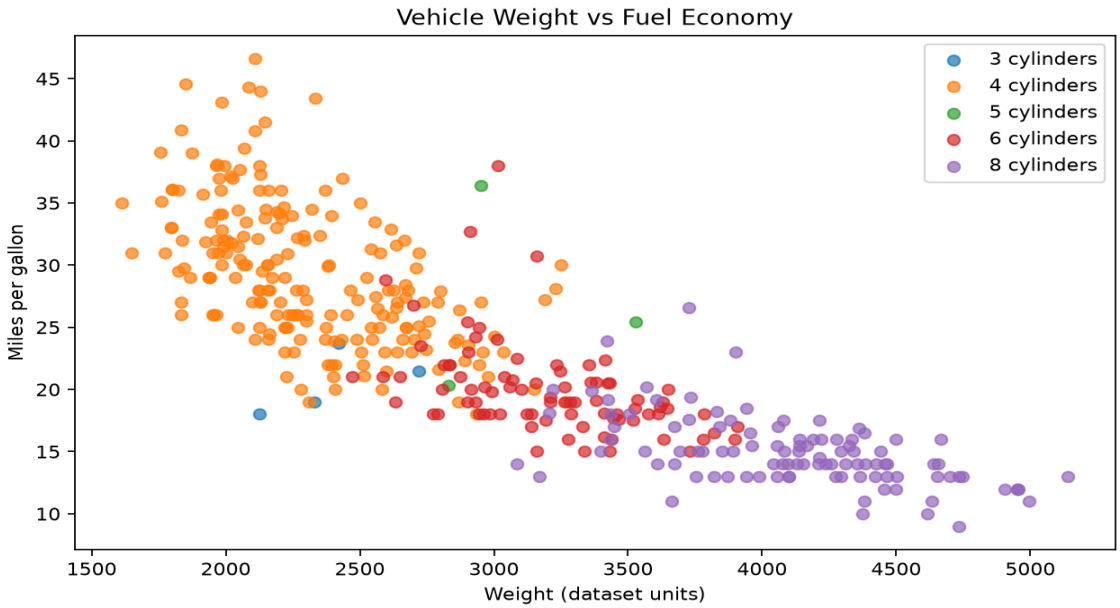


Observation: identify the most common cylinder configuration.

5. Visual EDA (continued)

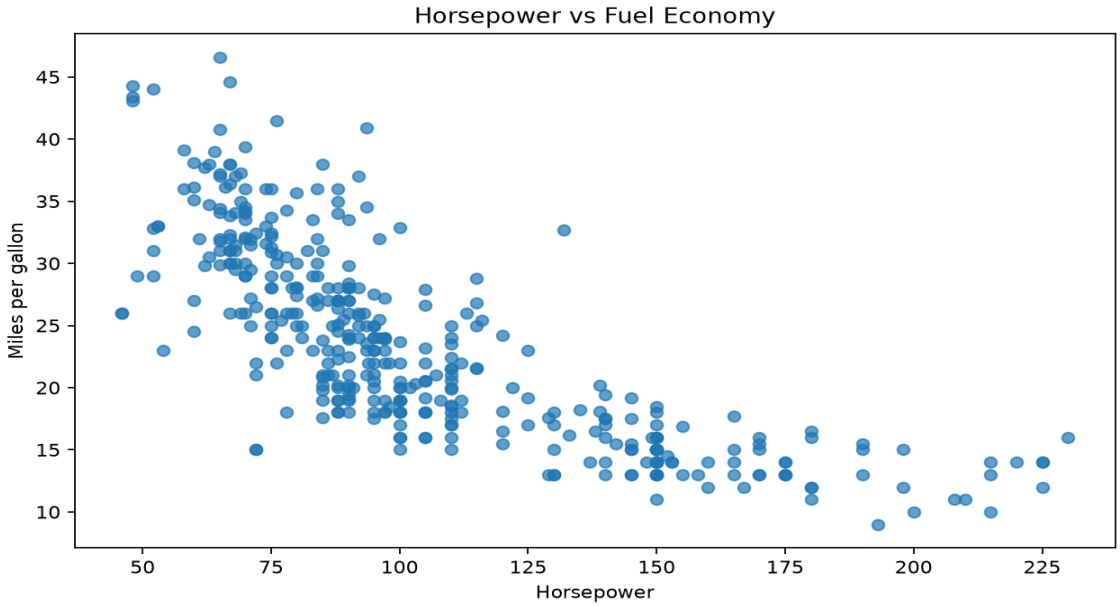


Observation: compare median MPG and spread across cylinder groups.

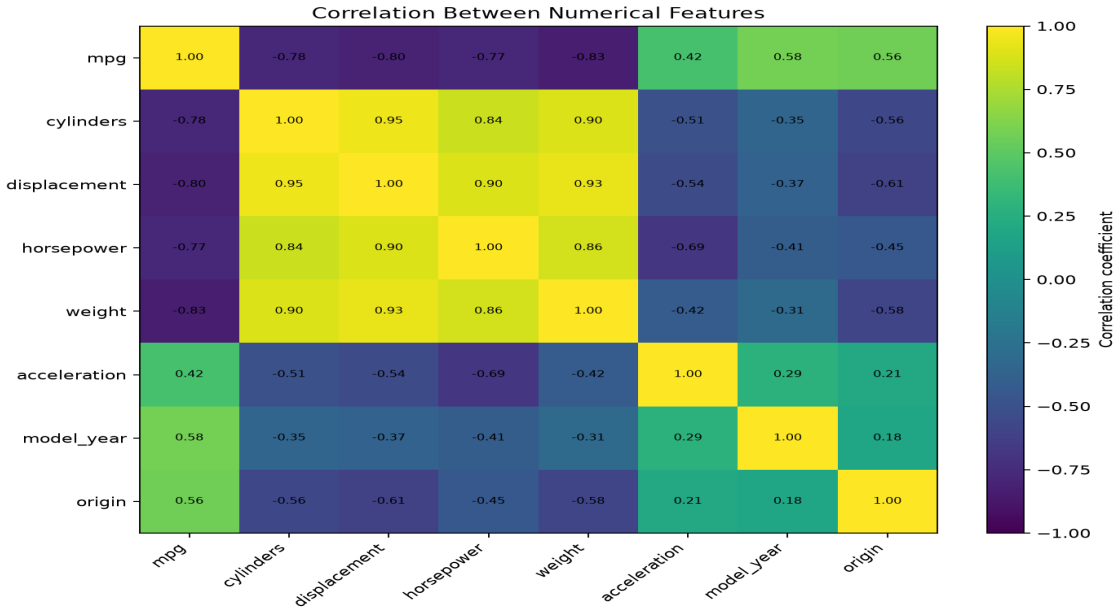


Observation: inspect the relationship between vehicle weight and MPG. Association does not prove causation.

5. Visual EDA (continued)



Observation: inspect whether MPG tends to change with horsepower.



Observation: compare the direction and strength of correlations among numerical features.

6. Key Findings

• The cleaned dataset contains 398 rows and 9 columns.

• Mean fuel economy is 23.51 MPG; median fuel economy is 23.00 MPG.

• The most common cylinder count is 4.

• The correlation between vehicle weight and MPG is -0.83.

• The correlation between horsepower and MPG is -0.77.

• Average MPG by cylinder count: 3 cylinders: 20.55 MPG; 4 cylinders: 29.29 MPG; 5 cylinders: 27.37 MPG; 6 cylinders: 19.99 MPG; 8 cylinders: 14.96 MPG.

• Potential outliers flagged by the IQR method (not automatically removed): mpg: 1, horsepower: 11, weight: 0, displacement: 0.

7. Limitations

The dataset represents historical vehicle models and does not contain car prices. Results are limited to the available attributes and should not automatically be generalized to all cars. Correlation does not establish causation.

8. Conclusion

The project demonstrates data inspection, cleaning, descriptive statistics, visual analysis and interpretation of automobile attributes. Future analysis could add a separately sourced price dataset or compare fuel economy across model years and origins.

References

UCI Machine Learning Repository, Auto MPG Dataset: https://archive.ics.uci.edu/dataset/9/auto+mpg